D 31730 **(Pages : 2)** **Name.....**

Reg. No.....

THIRD SEMESTER (CBCSS—UG) DEGREE EXAMINATION
NOVEMBER 2022

B.Com./B.B.A.

A 11—BASIC NUMERICAL METHODS

(2019 Admission onwards)

Time : Two Hours and a Half Maximum : 80 Marks

Part A

Answer **all** *questions.*

1. What is pure quadratic equation ?
2. What is meant by roots of the quadratic equation ?
3. What do you mean by the minor element of a matrix ?
4. What is triangular matrix ?
5. What is Skew Symmetric Matrix ?
6. Define harmonic progression.
7. What is meant by convergent series ?
8. What is Annuity ?
9. What is effective annual interest rate ?
10. What is Median ?
11. State any *two* merits of arithmetic mean.
12. Define quartile deviation.
13. Define Variance.
14. How do you calculate range and its co-efficient ?
15. What is positive skewness ?

(15 × 2 = 30, Maximum ceiling 25 marks)

Turn over

Part B

Answer **all** *questions.*

16. Solve $18 + 10x + 3x = 106 - 5x$.

17. Solve the equation $x^2 + 7x = 60$.

18. Find the determinant of $A = \begin{bmatrix} 4 & 1 & 2 \\ 1 & 2 & 5 \\ 2 & 7 & 8 \end{bmatrix}$.

19. Find the 10th term of Arithmetic progression : 4, 8, 12.....

20. Three numbers in ascending order in geometric progression such that their product is 729. Find the middle number.

21. Calculate simple interest and amount at end of the 7th year for Rs. 15,000 at 9 % per annum.

22. Find median for the values : 85, 57, 63, 9, 74, 34, 36, 93.

23. Calculate Arithmetic mean from the following data :

Values	:	10	20	30	40	50	60	70	80
Frequency	:	5	8	12	17	15	12	10	4

(8 × 5 = 40, Maximum ceiling 35 marks)

Part C

Answer any **two** *questions.*

24. Solve the following equations by using Cramer's rule :

$2x + 3y = 7$
$4x + 2y = 10$.

25. Find compound interest for Rs. 5,000 for 5 years if interest is payable annually at 5 % p.a

26. Find the 8th term and 9th term of the geometric progression 5, 10, 20....

27. Calculate Standard deviation and co-efficient of variation from the following values :

Size	:	2	4	6	8	10	12
Frequency	:	2	4	5	6	7	8

(2 × 10 = 20 marks)

D 31730-A **(Pages : 4)** **Name.....**

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A 11—BASIC NUMERICAL METHODS

(2019 Admission onwards)

(Multiple Choice Questions for SDE Candidates)

Time : 15 Minutes	**Total No. of Questions : 20**	**Maximum : 20 Marks**

INSTRUCTIONS TO THE CANDIDATE

1. This Question Paper carries Multiple Choice Questions from 1 to 20.
2. The candidate should check that the question paper supplied to him/her contains all the 20 questions in serial order.
3. Each question is provided with choices (A), (B), (C) and (D) having one correct answer. Choose the correct answer and enter it in the main answer-book.
4. The MCQ question paper will be supplied after the completion of the descriptive examination.

A 11—BASIC NUMERICAL METHODS

(Multiple Choice Questions for SDE Candidates)

1. One equation $ax^2 + 4 = 0$ ·

   (A) Quadratic equation. (B) Cubic equation.

   (C) Simple equation. (D) None of these.

2. One roots of the equation $3x^2 - 1 = 0$ are ————————.

   (A) Irrational. (B) Imaginary.

   (C) Rational. (D) Integers.

3. One A.M. of a and b is ————————.

   (A) ab. (B) $\frac{a+b}{2}$.

   (C) $a + b$. (D) $\frac{ab}{2}$.

4. One common difference of the A.P. 1, –1, –3, –5, ———— is :

   (A) 1. (B) –1.

   (C) – 2. (D) 2.

5. If the sum of two numbers is 8 and their product is 15, numbers are :

   (A) – 5, – 3. (B) 5, 3.

   (C) 5, – 3. (D) 2, 6.

6. Find the 7th term of the series 1, 4, 7, ————————.

   (A) 22. (B) 19.

   (C) 16. (D) 25.

7. One value of determinant $\begin{vmatrix} a & b \\ c & d \end{vmatrix}$ is :

   (A) $ab - cd$. (B) $ac - bd$.

   (C) $ab - bc$. (D) $ad - bc$.

8. A square matrix in which all the elements except those in leading diagonal are zero is called ———— matrix.

(A) Zero. (B) Null.

(C) Diagonal. (D) None.

9. Simple interest for the sum of Rs. 3,000 at 7 p.a. for 3 years.

(A) 420. (B) 630.

(C) 6,300. (D) 63.

10. Statistical result are :

(A) Absolutely correct. (B) Not true.

(C) True on average. (D) Universally true.

11. An attribute is :

(A) A qualitative characteristic. (B) A quantitative.

(C) Both. (D) None of these.

12. The classification 10–20, 20-30 is the example of :

(A) Exclusive. (B) Inclusive.

(C) Both. (D) None.

13. To find the median, arrange the data in ————————.

(A) Ascending order. (B) Descending order.

(C) Ascending or Descending order. (D) No order.

14. ———————— is the reciprocal of the AM of reciprocal of observations.

(A) H.M. (B) G.M.

(C) A.M. (D) None.

15. Pie diagrams are ———————— dimensional diagrams.

(A) One. (B) Two.

(C) Three. (D) Four.

16. If the mid value of classes are 2.5, 7.5, 12.5, 17.5 the first class is :

(A) 2.5–7.5. (B) 2.5–2.5.

(C) 5–10. (D) 0–5.

17. Classification is one of the methods of ———— data.

(A) Presentation. (B) Analysis.

(C) Organising. (D) None.

18. Which measure of dispersion is the quickest to compute ?

(A) S.D. (B) Q.D.

(C) Range. (D) All.

19. When a frequency curve is more peaked than the normal curve, it is called ————.

(A) Platy kurtic. (B) Lepto Kurtic.

(C) Meso Kurtic. (D) None.

20. Statistics are :

(A) Aggregate of facts. (B) Numerically expressed.

(C) Systematically collected. (D) All these.